Section Report 4: *Pseudomonas* Transcriptome

Purpose:
The purpose of this report is to employ the use of transcriptomics to explore and compare the differences between upregulated and downregulated genes from both wild type and mutant strains of a *Pseudomonas* sequence.

Methods:

1. Import the data as new history
    a. Starting point of study
2. Falco (Galaxy Version 1.3.0+galaxy0)
    a. Performs high-speed quality control by analyzing raw FASTQ files to identify sequencing errors, bias, and adapter contamination.
3. FastP (Galaxy Version 1.3.3+galaxy0)
    a. Performs quality control and preprocessing on FASTQ data to help streamline the steps for next-gen sequencing.
4. HISAT2 (Galaxy Version 2.2.2+galaxy0)
    a. Maps and aligns next-gen sequencing reads against a reference genome.
5. StringTie (Galaxy Version 2.2.3+galaxy0)
    a. Acts as a high-performance transcriptome assembler and quantifier die RNA-Seq data.
6. StringTie Merge (Galaxy Version 2.2.3+galaxy0)
    a. Combines multiple sets of assembled transcripts from different RNA-Seq samples into a single, unified, and non-redundant transcriptome.
7. GFF Compare (Galaxy Version 0.12.10+galaxy0)
    a. Used to systematically compare, merge, annotate, and evaluate the accuracy of transcript assemblies against a reference annotation.
8. FeatureCounts (Galaxy Version 2.1.1+galaxy0)
    a. An ultrafast, high efficient read quantification program used to count mapped sequencing reades against known genomic features.
9. Unlocking the Hidden Stuff
    a.
10. DESeq2 (Galaxy Version 2.11.40.8+galaxy2)
    a. Performs differential gene expression analysis on a high-throughput sequencing data. Helps identify which genes are statistically significantly upregulated or downregulated.
11. Filter (Galaxy Version 1.1.1)
    a. Restricts a tabular dataset by selecting only the rows that satisfy a user-defined conditional statement
12. Adding new header to Galaxy
    a.
13. Concatenate multiple datasets or collections (Galaxy Version 1.0.0)
    a. Combines disparate data inputs into a single output or a collection.
14. Volcano Plot (Galaxy Version 4.0.3+galaxy0)
    a. Allows for visualization of differential expression results from high-dimensional biological datasets.

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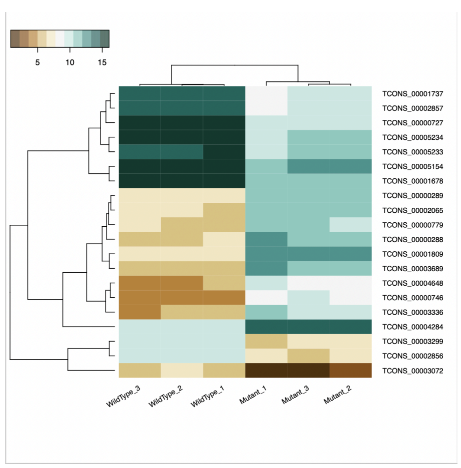
15. Excel Stuff
    a.
16. Join two datasets (Galaxy Version 2.1.3)
    a. Combines lines from two separate tabular datasets into a single output by matching values in a common field
17. Cut (Galaxy Version 1.0.2)
    a. Extracts specific columns from a tabular dataset and rearranges the columns in the output.
18. Heatmap2 (Galaxy Version 3.3.0+galaxy0)
    a. Generates a visual heatmap to display data such as gene expression in numerical format.
19. Filter (Galaxy Version 1.1.1)
    a. Removes rows that don't meet user-defined criteria by restricting datasets by applying conditional logic to specific columns.
20. Convert (Galaxy Version 1.0.1)
    a. Reformats bioinformatics data so that it becomes compatible with downstream analysis software
21. Replace Text (Galaxy Version 9.5+galaxy3)
    a. Manipulates tabular data to perform automated pattern-based text replacement in order to clean data, standardize identifiers, or prep files for downstream statistical analysis.
22. Cut (Galaxy Version 1.0.2)
    a. Extracts and rearranges specific columns from a tabular data.
23. Join two datasets (Galaxy Version 2.1.3)
    a. Based on a shared value in a specific column, it merges two tabular files side-by-side.

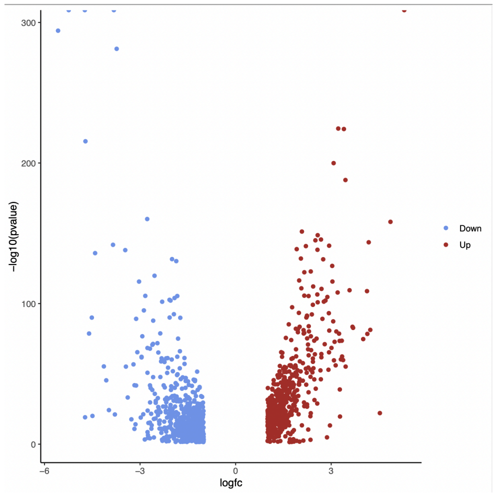
Results:

- Heatmap



- Interpretation: In our heatmap, the darker the teal, the more expressed the gene is while the darker the brown, the less expressed - white is the middle of the spectrum. For the most part, it seems like the top half and the bottom half of the graph show the most up & down regulated expression.

- Volcanoplot



  - Interpretation: Genes being upregulated will be on the right/positive (red) side of the graph while genes that are being downregulated will be on the left/negative (blue) side of the graph. Also, the higher the points are, the more significance they carry. We can see that on both ends of the graph, we have dots being cut off. This is because they go past the bounds of the program, meaning they are the most altered points.